AWS Cloud Infrastructure Lab

EBS Storage, Snapshots and Data Recovery

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Lab: Working with Amazon EBS

1. Project Overview

In this lab, I worked with Amazon Elastic Block Store (EBS) to add storage to an existing EC2 instance. I connected to the Linux server, set up the new disk, saved a test file and created a snapshot. I then deleted the file from the original volume and used the snapshot to create another volume. The final step was checking whether the file could be accessed from the restored volume.

2. Objectives

- Create a 1 GiB EBS volume in the same Availability Zone as the EC2 instance.
- Attach the volume and configure an ext3 file system in Linux.
- Mount the storage and create a test file.
- Create an EBS snapshot and restore it as a new volume.
- Confirm that the test file is present on the restored volume.

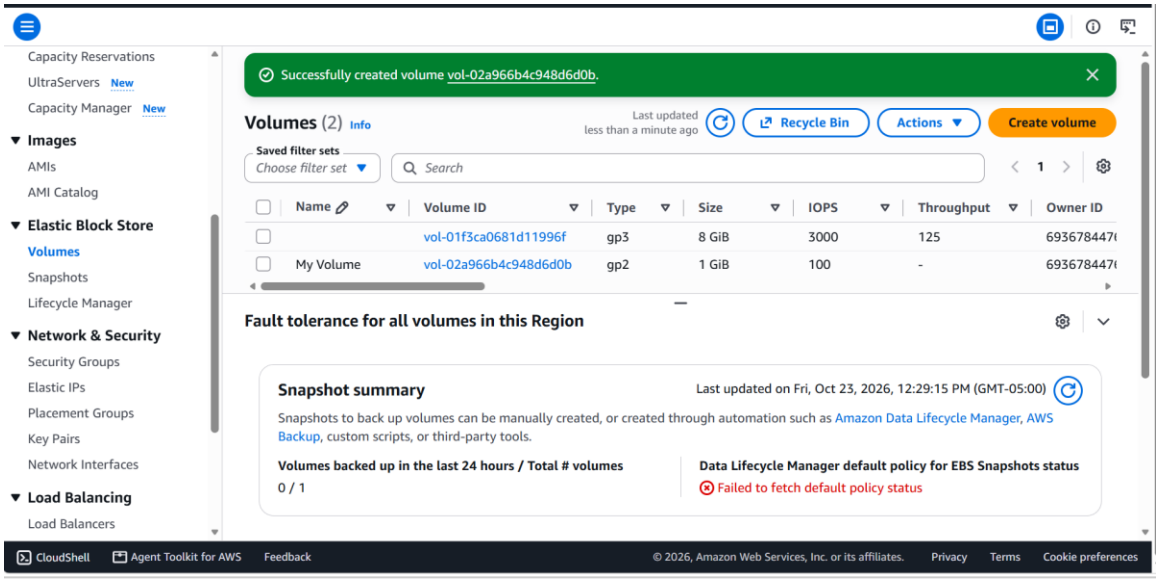
3. Lab Environment

Cloud platform	**Amazon Web Services (AWS) – training lab**
Services	Amazon EC2, Amazon EBS, AWS Systems Manager Session Manager
Operating system	Linux (EC2 lab instance)
Original additional volume	1 GiB EBS volume, gp2 as specified in the lab
File system	ext3
Mount points	/mnt/data-store and /mnt/data-store2

4. Work Completed

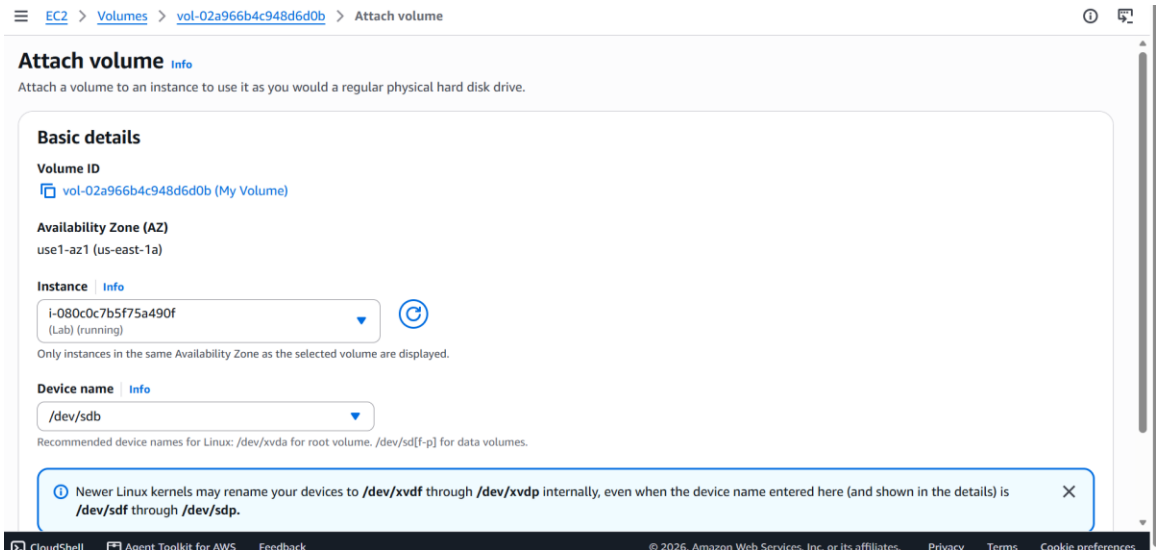
Step A – Created an EBS volume

I opened the EC2 console and created a 1 GiB EBS volume named My Volume. I selected the same Availability Zone as the existing Lab instance, since the volume needed to be attached to that instance.



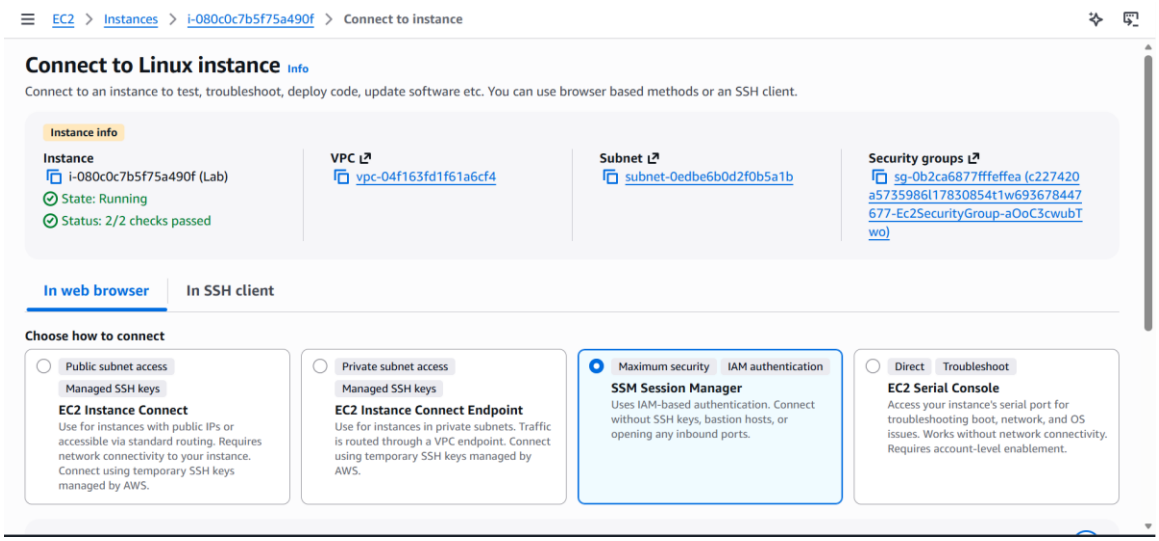
Step B – Attached the volume

I attached My Volume to the Lab EC2 instance. After attachment, the volume status changed to In-use.



Step C – Configured the storage in Linux

I connected to the instance using Session Manager. I checked the existing storage, created an ext3 file system on the new device, made the /mnt/data-store directory and mounted the volume there. I also updated /etc/fstab as directed in the lab so the volume could be mounted at startup.



Step D – Tested file storage

I created file.txt on the new volume and checked its contents in the terminal. This confirmed that I could write to and read from the mounted storage.

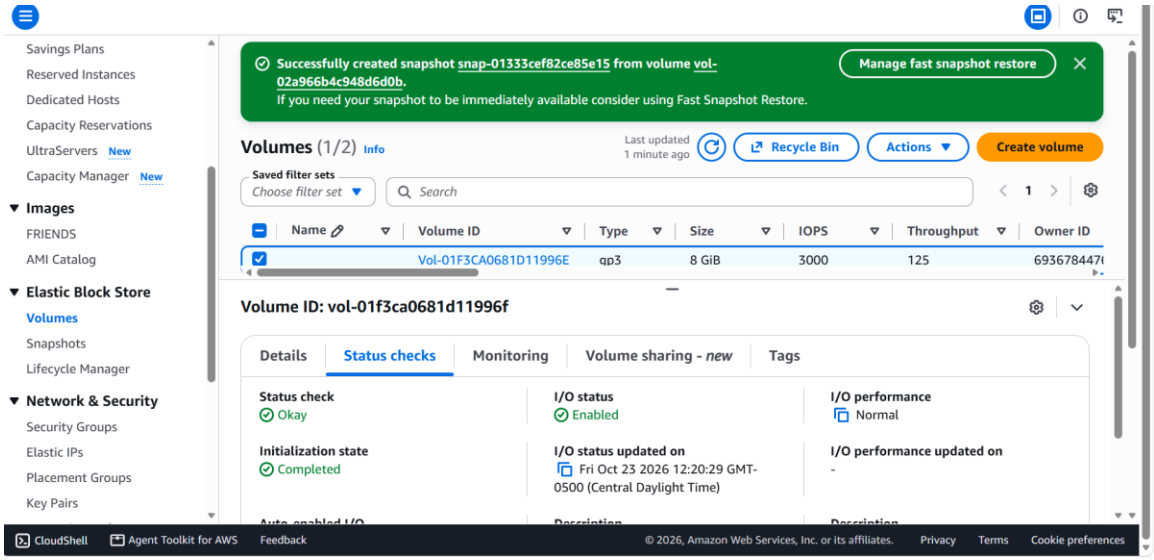
```
tmpfs           96M     0   96M   0% /run/user/0
[ec2-user@ip-10-1-11-253 ~]$ sudo mkfs -t ext3 /dev/sdb
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: d978d382-a0fb-465d-8146-328f4e31e7a3
Superblock backups stored on blocks:
       32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-1-11-253 ~]$ sudo mkdir /mnt/data-store
[ec2-user@ip-10-1-11-253 ~]$ sudo mount /dev/sdb /mnt/data-store
[ec2-user@ip-10-1-11-253 ~]$ echo "/dev/sdb   /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab
/dev/sdb   /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-253 ~]$ cat /etc/fstab
#
UUID=242449b2-b330-47d6-88c0-187dac3594f0     /           xfs    defaults,noatime  1   1
UUID=A2E8-3E76        /boot/efi       vfat    defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
/dev/sdb   /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-253 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  452K  192M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           479M     0  479M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            96M     0   96M   0% /run/user/0
/dev/xvdb       975M   60K  924M   1% /mnt/data-store
[ec2-user@ip-10-1-11-253 ~]$
```

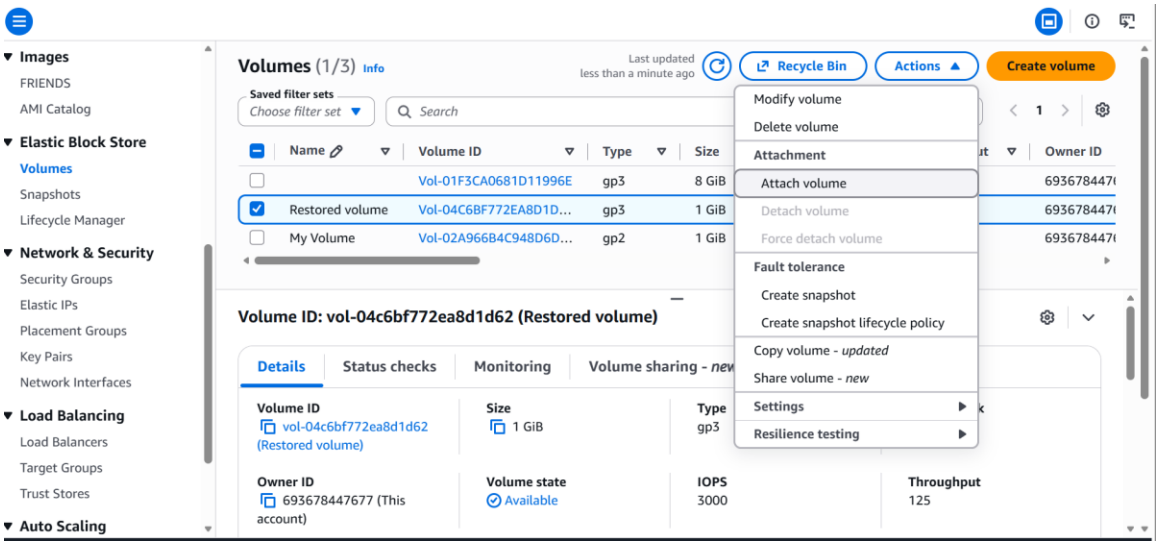
Step E – Created a snapshot

In the EC2 console, I created an EBS snapshot named My Snapshot and waited for its status to show Completed. After the snapshot was created, I deleted the test file from the original mounted volume.



Step F – Restored the snapshot

I created a new EBS volume from My Snapshot, named it Restored Volume and attached it to the same EC2 instance. I mounted it under /mnt/data-store2.



Step G – Verified recovery

I checked the restored mount point for file.txt. Finding the file on the new volume showed that the earlier snapshot contained the data, even though I had deleted the file from the original volume afterward.

5. Linux Commands Used

These are the main commands used during the lab. The device paths below follow the lab instructions; the actual device name should be checked with lsblk -f before formatting or mounting.

df -h

lsblk -f

sudo mkfs -t ext3 /dev/sdb

```
sudo mkdir /mnt/data-store

sudo mount /dev/sdb /mnt/data-store

cat /mnt/data-store/file.txt

sudo mkdir /mnt/data-store2

sudo mount /dev/sdc /mnt/data-store2

ls /mnt/data-store2/

cat /mnt/data-store2/file.txt
```

6. Result

The main result of this lab was recovering the test file from a volume created using an EBS snapshot. This showed me how snapshots can be used to bring back data from an earlier point in time. The screenshots included with this report document the configuration and verification steps.

7. What I Learned

Before this lab, I knew that cloud instances use storage, but I had not worked through the full process of adding a separate EBS volume and restoring it from a snapshot. I got more practice with the AWS console and Linux commands such as df, mount and cat. I also learned that taking a snapshot is only part of a backup process: it is important to test whether the data can actually be restored.

8. Notes

This project was completed in a temporary AWS training environment. The screenshots are my evidence of the hands-on work. The lab steps were provided as part of the course; this report is my own summary of the work.